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Short Communication

Misinformation on resuscitation and first aid as an uncontrolled problem that demands close attention: a brief scoping review

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ABSTRACT

Objectives: Misinformation is currently recognised by the World Health Organization as an apparent threat to public health. This study aimed to provide an outline of published evidence on misinformation related to the potentially life-saving interventions – first aid and cardiopulmonary resuscitation (CPR).
Study design: A scoping review.
Methods: The review was conducted in accordance with the PRISMA Extension for Scoping Reviews. English-language publications describing original studies that evaluated the quality of publicly available information on first aid and/or CPR were included without limitations to the year of publication.
Results: Forty-four original studies published between 1982 and 2023 were reviewed. Annual number of publications varied from 0 to 6. The studies have focused on the evaluation of information concerning initial care of cardiac arrest, choking, heart attack, poisoning, burns, and other emergencies. Forty three studies (97.7 %) have reported varying frequencies of misinformation, when public sources, including websites, YouTube videos, and modern artificial intelligence-based chatbots, omitted life-saving instructions on first aid or CPR or contained incorrect information that contradicted relevant international guidelines. Eleven studies (25.0 %) have also revealed potentially harmful advice, which, if followed by an unsuspecting person, may cause direct injury or death of a victim.
Conclusions: Misinformation concerning CPR and first aid cannot be ignored and demands close attention from relevant stakeholders to mitigate its harmful impacts. More studies are urgently needed to determine optimal methods for detecting and measuring misinformation, to understand mechanisms that drive its spread, and to develop effective measures to correct and prevent misinformation.
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Introduction

With intensively growing public access to internet and mobile technologies, the global information ecosystem undergoes unprecedented and drastic changes. Search for health information currently follows a decentralised model where people shift from traditional sources, such as TV or print, to more digital, mobile, and social platform-dominated media environments.[1] This causes an overabundance of information, including the dissemination of false or misleading openly available content that contradicts with the best available scientific evidence, and therefore may impact public health through risk-taking behavioural choices or inaction.

Major health stakeholders, including the World Health Organization, recognise misinformation as an apparent threat to public health.[2] Increased attention to the problem has potentiated extensive research that helps to better understand, measure and respond to misinformation in various healthcare domains.[3] However, there is a lack of general comprehension of the extent of misinformation problem in relation to principles of initial care provided by lay public in emergencies, when inappropriate decision-making based on unreliable advice may have direct negative influence on health outcomes. Hence, a brief scoping review was conducted to outline cumulative published evidence on misinformation related to first aid and cardiopulmonary resuscitation (CPR).

Methods

The study was carried out in accordance with the PRISMA Extension for Scoping Reviews. The review protocol was not pre-registered.

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In October 2023, Pubmed was searched for English-language journal articles describing original studies that evaluated the quality of publicly available information on first aid and/or CPR (see Appendix for the search strategy). Due to the expected paucity of literature, research letters were included in analysis in addition to full articles, and no limitations to the year of publication were applied.

The search results were screened based on title and abstract, and potentially eligible records were exported into Zotero (Corporation for Digital Scholarship, Vienna, Virginia, USA) reference management software. Full texts were then reviewed by the author, and eligible articles were included in the final analysis. Using a pre-developed form, the following data were collected from each eligible article: authors' names, publication year, study aim, study object (e.g. websites or videos), studied health condition (e.g. burns or heart attack), study methods, main results, as well as an indication of whether the study detected insufficient, incorrect, inaccurate, or potentially harmful information related to CPR or first aid. The charted data were then analysed descriptively.

Results

The search produced a total of 1455 articles, of which 30 were identified as eligible. Fourteen additional articles were found when screening references. Ultimately, 44 original studies published between 1982 and 2023 were reviewed (see dataset[4]).

Annual number of publications varied from 0 to 6 with a slight trend towards increase in the recent years (Fig. 1). All but four studies were published by different first authors. Along with the small number of publications this suggests the lack of sustained research interest in the issue.

The studies have focused on the evaluation of information concerning initial care of cardiac arrest ($n = 22$, 50.0 %), poisoning ($n = 7$, 15.9 %), burns ($n = 5$, 11.4 %), dental trauma ($n = 2$, 4.5 %), choking ($n = 1$, 2.3 %), fever ($n = 1$, 2.3 %), heart attack ($n = 1$, 2.3 %), scalds ($n = 1$, 2.3 %), or a range of health emergencies (n = 4, 9.1 %). Analysed sources of information included online videos ($n = 14$, 31.8 %), websites ($n = 8$, 18.2 %), labels of commercial (household, hazardous) products ($n = 5$, 11.4 %), large language model-based chatbots ($n = 4$, 9.1 %), mobile apps ($n = 4$, 9.1 %), television dramas and movies ($n = 4$, 9.1 %), face-to-face ($n = 3$, 6.8 %) and online courses ($n = 1$, 2.3 %), textbooks and manuals ($n = 3$, 6.8 %), virtual voice-based conversational assistants ($n = 2$, 4.5 %), and a web search engine's question-answering system ($n = 1$, 2.3 %). The number of units of observation per study varied from 5 to 1459. The studies were highly heterogeneous in terms of design and applied methods of content evaluation.

As concerns the main results, all but one study ($n = 43$, 97.7 %) have reported varying frequencies of misinformation, when sources omitted potentially life-saving instructions on first aid or CPR (e.g. a recommendation to encourage the person to stop physical activity and to take their antianginal medication in case of suspected heart attack, or an instruction to immediately start resuscitation for unresponsive abnormally breathing person) or contained incorrect information that contradicted relevant international guidelines (like improper rate or depth of chest compressions for CPR, or misplacement of electrode pads for defibrillation). Furthermore, eleven studies (25.0 %) have revealed that various public sources of information, including websites, YouTube videos, and modern artificial intelligence-based question-answering systems, provided potentially harmful advice, which, if followed by an unsuspecting person, may cause direct injury or even death of a victim (for instance a recommendation to make the person vomit in case of toxic substance ingestion, to apply ice directly to acute burn wounds, or to use home remedies in chest pain). See dataset[4] for detailed information on each study.

Discussion

One could assume that inappropriate advice on first aid contained in open sources would not be translated into real-life application in a health emergency. In contrast to this view, the preliminary results of our ongoing survey among lay completers of an internet course on basic life support ($n = 165$) showed that 13.9 % of respondents have ever searched for brief instructions on first aid online in true emergency situations, and another 8.5 % reported such experience from their kith. Also, a recent study that

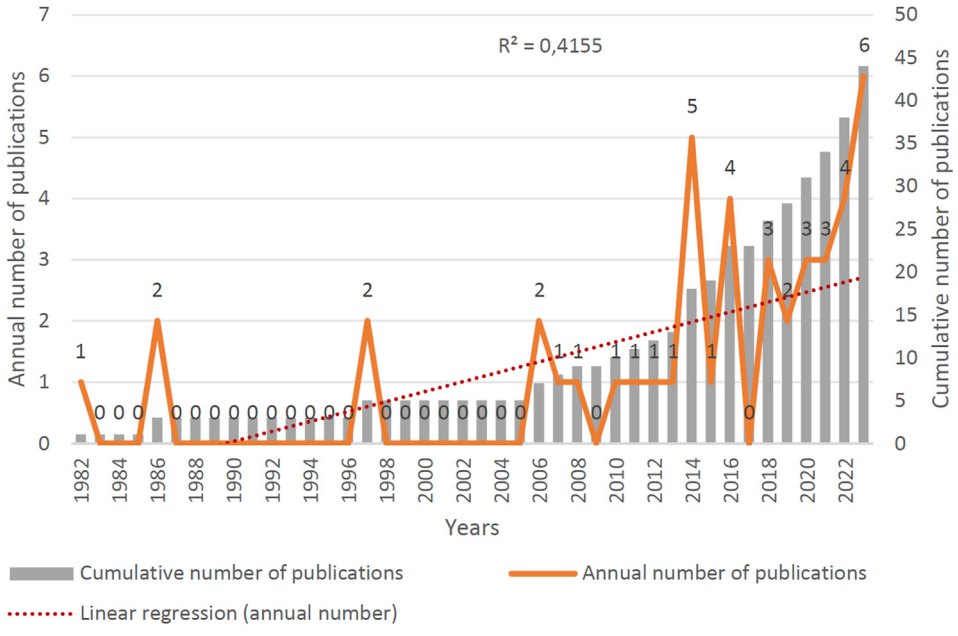


Fig. 1. Quantitative distribution of the reviewed publications.

analysed public comments on YouTube first aid-related videos has revealed evidence of laypeople's real-life efforts to get immediate assistance and advice on first aid online during a sudden health crisis.[5]

Taken together with the scoping review results, this indicates that misinformation concerning CPR and first aid constitutes a major public health problem that cannot be ignored and demands close attention from relevant stakeholders to mitigate its harmful impacts.

That said, insufficient regard is currently given to this issue. Whereas international guidelines on CPR and first aid place an emphasis on emerging digital technologies, including social media and smartphone apps, as highly perspective means for engaging the community in education and provision of emergency care,[6,7] little to no attention has been paid in these reputable documents towards the risks of uncontrolled distribution of incomplete, inaccurate or potentially harmful information, and strategies to tackle the problem.

Misinformation can be very quickly propagated through social media,[8] which are increasingly popular nowadays. Furthermore, recent appearance of publicly available artificial intelligence chatbots, which can generate vast amounts of convincing human-like text, could leverage misinformation spread at an outstanding scale causing an "AI-driven infodemic."[9] Therefore, prompt interventions to control misinformation are required.

In accordance with its mission to promote implementation of evidence-based lifesaving practices, the International Liaison Committee on Resuscitation presumably should take the lead in developing a unified organisational framework to counter the spread of misinformation on first aid and resuscitation. The misinformation problem should be called into attention of resuscitation research community and incorporated into the global research agenda on CPR and first aid. More studies are urgently needed to determine optimal methods for detecting and measuring misinformation, to understand mechanisms that drive its spread, to investigate the impact of misinformation on health outcomes, and to develop effective measures to correct and prevent misinformation. As the studies on misinformation are known to be of low quality[3] and heterogeneous in methodologies, the development of recommendations on study design and reporting, as well as the creation of standardised instruments for content evaluation, would facilitate the research.

Author statements

Ethical approval

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Competing interest

None declared.

Data statement

All data that support the findings of this study are openly available in Mendeley Data repository.[4]

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.puhe.2024.01.005.

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